

GUARDRAIL LENGTH-OF-NEED CALCULATIONS

Bridge approach guardrail based on MDOT Roadway Design Manual Table 9-6-A and GR-4 dimensions.

Project	SR 8	Job Number	(not provided)
Route / Location	(not provided)	Calculation Date	2026-06-22
Prepared By	(not provided)	Checked By	(not provided)

DESIGN INPUTS

Facility	Two-Lane, Two-Way Roadway	Design Speed	65 mph
Design ADT	6000	Runout Length, LR	290.0000 ft
Hazard Offset, LA	30.0000 ft	Near-Side Offset, L2	10.0000 ft
Opposing-Side Offset, L2	22.0000 ft	Lane Width	12.0000 ft
Transition Length, L1	18.1458 (18'-1.7500")	Terminal Section	25.0000 ft
Flare Rate, a/b	30/1	Guardrail Increment	12.5000 ft

FINAL DESIGN OUTPUTS

A - Near Side	B - Near Rail	C - Opposing Side	D - Opposing Rail
155.6458 ft	112.5000 ft	68.1458 ft	25.0000 ft
Near Side Including Gating	168.1458 ft	Opposing Side Including Gating	80.6458 ft

SUPPORTING RESULTS

Near Minimum X	150.6406 ft	Near Required Rail, B	107.4947 ft
Opposing Minimum X	62.9095 ft	Opposing Required Rail, D	19.7637 ft
Rounded Rail, B	112.5000 ft	Rounded Rail, D	25.0000 ft

Design basis: MDOT Roadway Design Manual, Table 9-6-A, Barrier Length of Need. Calculated guardrail lengths are rounded in accordance with the existing 12.5000-foot increment workflow.

Engineering disclaimer: This calculation package documents user-entered inputs and computed results. It is not a sealed or stamped engineering document and must be independently reviewed for project applicability.

DETAILED CALCULATIONS

Project: SR 8 | Job Number: (not provided)

COMMON PARAMETERS

Design Speed	65 mph	Design ADT	6000
LA	30.0000 ft	L1	18.1458 (18'-1.7500")
Terminal Section	25.0000 ft	LR	290.0000 ft

NEAR SIDE - OUTPUTS A AND B

Near-Side Length of Need

Minimum X	150.6406 ft	Required Rail Length	107.4947 ft
Rounded Rail Length	112.5000 ft	Final Overall Length	155.6458 ft

Length-of-need equation and substitution

$$X = (30.0000 + 0.0333 \times 18.1458 - 10.0000) / (0.0333 + (30.0000/290.0000))$$

B Required	B(required) = 150.6406 - 18.1458 - 25.0000 = 107.4947	107.4947 ft
Output B	B = round up max(12.5000, 107.4947) to 12.5000-ft increment = 112.5000	112.5000 ft
Output A	A = 112.5000 + 18.1458 + 25.0000 = 155.6458	155.6458 ft

OPPOSING SIDE - OUTPUTS C AND D

Opposing-Side Length of Need

Minimum X	62.9095 ft	Required Rail Length	19.7637 ft
Rounded Rail Length	25.0000 ft	Final Overall Length	68.1458 ft

Length-of-need equation and substitution

$$X = (30.0000 + 0.0333 \times 18.1458 - 22.0000) / (0.0333 + (30.0000/290.0000))$$

D Required	D(required) = 62.9095 - 18.1458 - 25.0000 = 19.7637	19.7637 ft
Output D	D = round up max(12.5000, 19.7637) to 12.5000-ft increment = 25.0000	25.0000 ft
Output C	C = 25.0000 + 18.1458 + 25.0000 = 68.1458	68.1458 ft

All displayed calculated lengths are shown to four decimal places. Internal calculation precision and the established 12.5000-foot increment rounding method are unchanged.

APPENDIX A

MDOT Roadway Design Manual Reference

Embedded reference document included for calculation traceability.

Table 9-6-A
BARRIER LENGTH OF NEED

Design Speed (mph)	Design Traffic Volume (ADT)				Shy Line Offset (ft)	Steepest Flare Rate for Barrier Inside Shy Line	Steepest Flare Rate for Barrier Beyond Shy Line	
	Over 10,000	5000-10,000	1000-5000	Under 1000			Rigid (Concrete Barrier)	Semi-Rigid or Flexible (W Beam, Thrie Beam)
	Runout Length L _R (ft)	Runout Length L _R (ft)	Runout Length L _R (ft)	Runout Length L _R (ft)				
70	360	330	290	250	9.0	30:1	20:1	15:1
65	330	290	250	225	8.5	28:1	19:1	15:1
60	300	250	210	200	8.0	26:1	18:1	14:1
55	265	220	185	175	7.0	24:1	16:1	12:1
50	230	190	160	150	6.5	21:1	14:1	11:1
45	195	160	135	125	6.0	18:1	12:1	10:1
40	160	130	110	100	5.0	16:1	10:1	8:1
35	135	110	95	85	4.5	15:1	9:1	8:1
30	110	90	80	70	4.0	13:1	8:1	7:1

Equations to determine end of barrier need:

Flared Design

$$X = \frac{L_A + \frac{b}{a}(L_1) - L_2}{\frac{b}{a} + \frac{L_A}{L_R}}$$

$$Y = L_A - \frac{L_A}{L_R}(X)$$

Non-flared Design

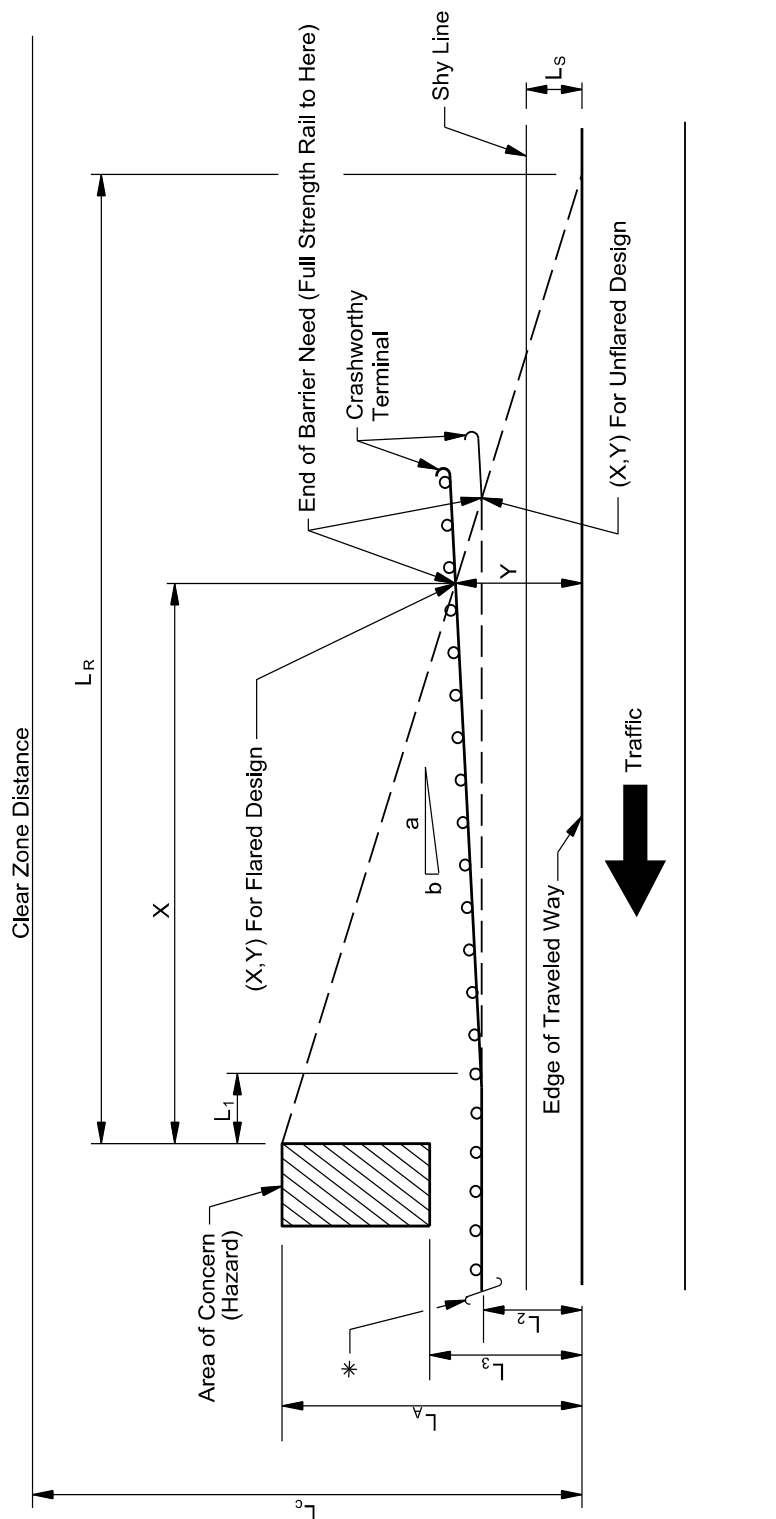
$$X = \frac{L_R (L_A - L_2)}{L_A}$$

$$Y = L_2$$

Definition of Terms:

X = length of need in advance of the area of concern
Y = lateral offset from edge of traveled way
b/a = barrier flare (e.g., 1/30)
L_c = recommended clear zone (feet)
L_A = distance from edge of traveled way to back of hazard (feet)
L_s = shy line offset, or distance at which barrier is no longer perceived as a hazard by a driver (feet)

L_R = runout length (feet)
L₁ = distance from hazard to where barrier flare begins (feet)
L₂ = distance from edge of traveled way to barrier (feet)
L₃ = distance from edge of traveled way to front of hazard (feet). (L₃ - L₂) should equal or exceed deflection distance.



BARRIER LENGTH OF NEED
Figure 9-6-A

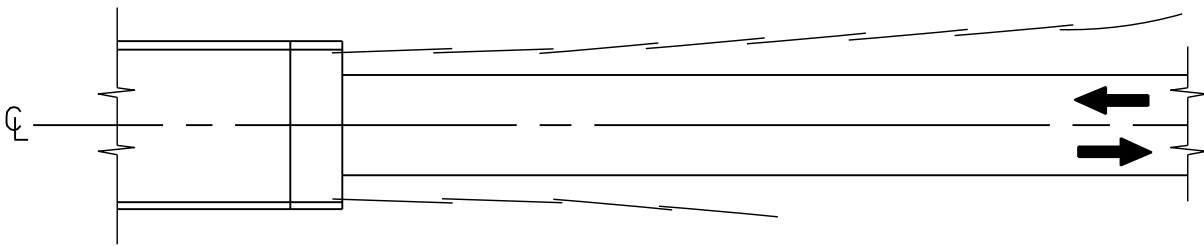
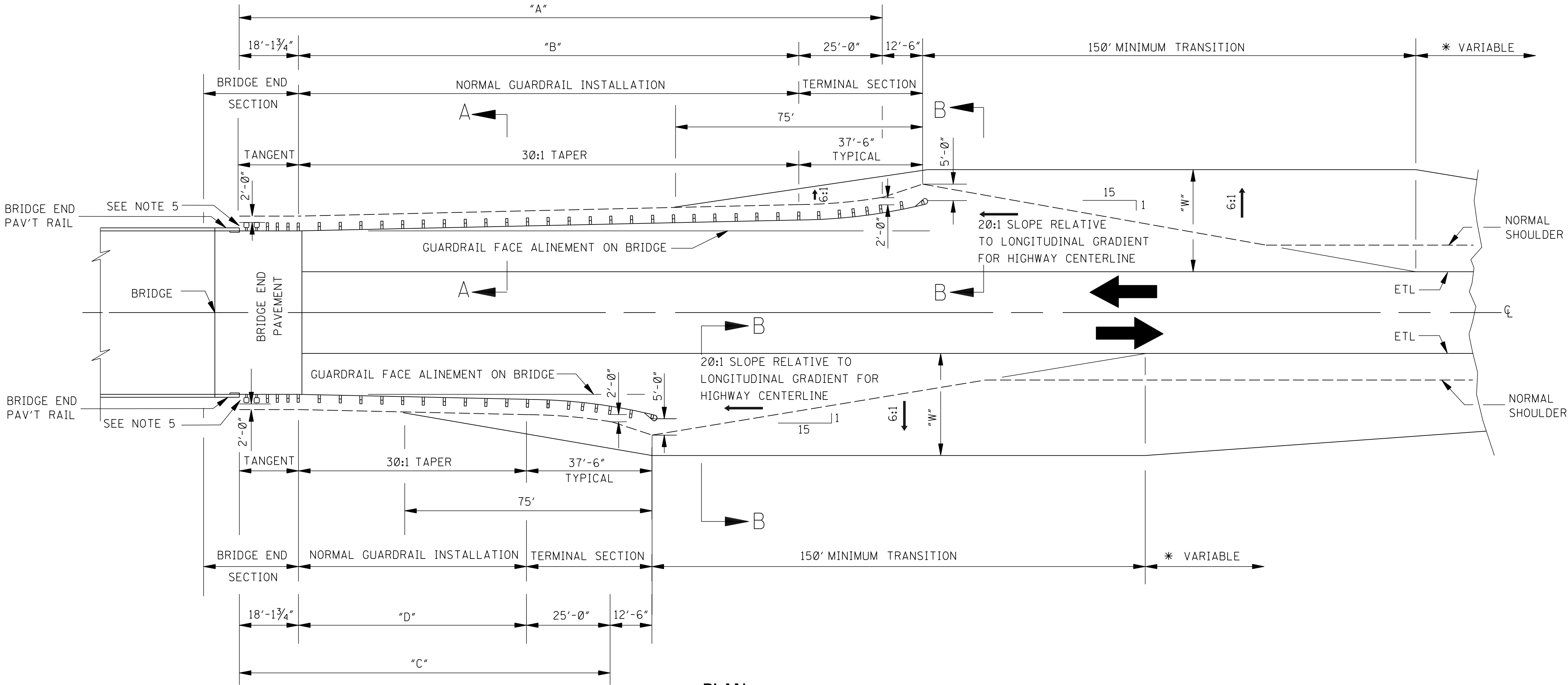
* The distance beyond the hazard should be determined by a length-of-need calculation for opposing traffic, if applicable.

APPENDIX B

MDOT Standard Plan GR-4a - Two-Lane, Two-Way Highways

Embedded reference document included for calculation traceability.

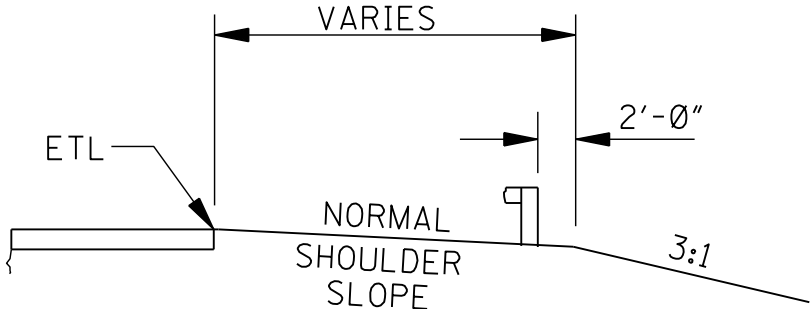
* NOTE: IF FORESLOPE, SHOWN ELSEWHERE ON PLANS, IS OTHER THAN 6:1, TRANSITION WILL OCCUR IN AREA SHOWN.



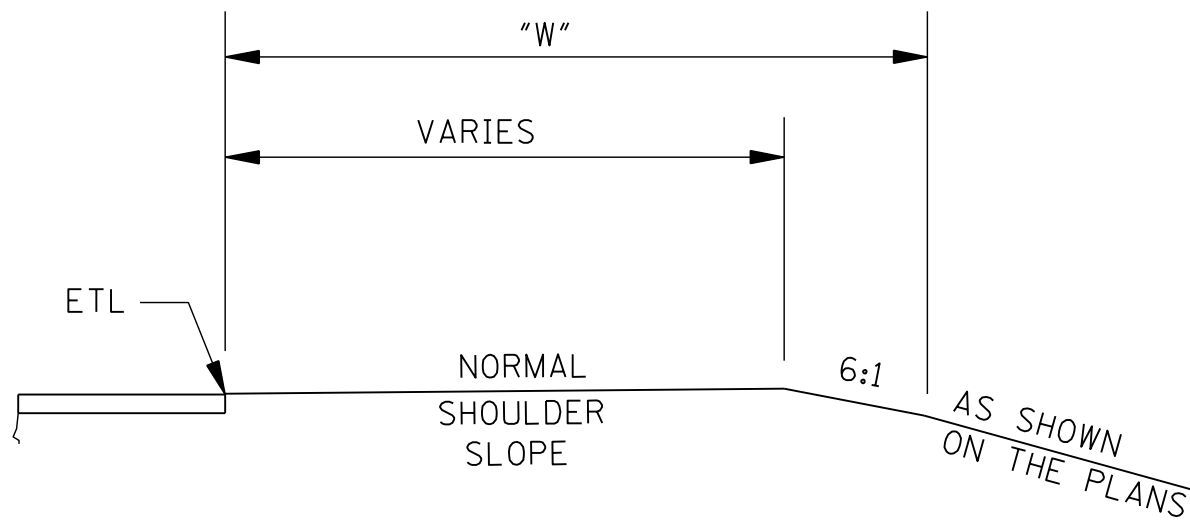
DETAIL OF GUARDRAIL SECTION LAPS

GENERAL NOTES:

- VALUES FOR "A", "B", "C" AND "D" WILL BE SHOWN ELSEWHERE ON THE PLANS.
- FOR DETAILS PERTINENT TO INSTALLATION OF THE TERMINAL SECTION, SEE MANUFACTURER'S SPECIFICATIONS AND DRAWINGS OR ELSEWHERE ON PLANS
- GUARDRAIL SECTIONS SHALL BE LAPPED IN THE DIRECTION OF TRAFFIC APPROACHING THE BRIDGE.
- THE OVERALL LENGTH OF GUARDRAIL IS MEASURED FROM THE CONNECTING END ON THE BRIDGE.
- IN THE ABSENCE OF A BRIDGE END PAVEMENT RAIL, CONNECT THE BRIDGE END SECTION TO THE BRIDGE RAIL (SEE WK. NOS. GR-2 THRU GR-2G). THE SHOULDER WIDTH AT THE BRIDGE END PAVEMENT RAIL OR BRIDGE END RAIL SHOULD BE SUFFICIENTLY WIDE TO PROVIDE A MINIMUM OF 2'-0" BEHIND THE BACK OF POST BEFORE THE SLOPE BREAK (HINGEPOINT).
- TYPE, DETAILS AND LIMITS OF GUARDRAIL BRIDGE END SECTION WILL BE SHOWN ELSEWHERE ON THE PLANS.
- W = SHOULDER WIDTH + FORESLOPE WIDTH. DIMENSIONS FOUND ELSEWHERE ON THE PLANS.



SECTION A-A



SECTION B-B

				BY	MISSISSIPPI DEPARTMENT OF TRANSPORTATION ROADWAY DESIGN DIVISION STANDARD PLAN
				REVISION	GUARDRAIL: TYPICAL INSTALLATION AT BRIDGE APPROACHES FOR 2-LANE, 2-WAY HIGHWAY
				DATE	ISSUE DATE: AUGUST 01, 2017
					MDOT MISSISSIPPI DEPARTMENT OF TRANSPORTATION WORKING NUMBER GR-4A SHEET NUMBER 6215